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Eduardo França Porto

CropTrack

SÃO PAULO
2025

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Eduardo França Porto

CropTrack

Final Course Project submitted to the Institute of Technology and Leadership (INTELI), to obtain a bachelor's degree in Computer Engineering

Advisor: Prof. Rodrigo Nicola

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[text in which the author presents a quotation, followed by an indication of authorship, related to the subject matter discussed in the body of the work]

Resumo **(mandatory item – NBR 14724, item 4.2.1.7)**

Porto, Eduardo. CropTrack:. 2025. nº de folhas xxx. TCC (Graduação) – Curso Engenharia de Hardware, Instituto de Tecnologia e Liderança, São Paulo, 2025.

Este trabalho apresenta o CropTrack, uma solução de agricultura de precisão voltada para a detecção precoce de doenças em lavouras de café utilizando visão computacional. O monitoramento tradicional de lavouras é manual, custoso e propenso a falhas, resultando em perdas significativas de produtividade. O objetivo deste projeto foi desenvolver e validar uma plataforma SaaS que integra um modelo de Deep Learning para classificação binária (saudável vs. doente) de folhas de café. A metodologia envolveu o desenvolvimento de modelos CNN (com destaque para a arquitetura BinaryCNN_Deep que atingiu 90-95% de acurácia), integrados a uma aplicação web com interface de mapas para gestão de talhões. Os resultados demonstraram que a solução é capaz de identificar anomalias com alta precisão, oferecendo aos produtores uma ferramenta eficaz para tomada de decisão rápida e redução de desperdícios.

Palavras-Chave: Visão Computacional; Café; Agricultura de Precisão; Deep Learning; Monitoramento de Lavouras.

ABSTRACT (mandatory item – NBR 14724, item 4.2.1.8)

Porto, Eduardo. CropTrack, 2025. nº of pages xxx . Final course project (Bachelor) – Course Hardware Engineering, Institute of Technology and [Leadership , São Paulo, 2025.] **(The reference is an optional item. NBR 6028, item 4.1.6)**

[This work presents CropTrack, a precision agriculture solution aimed at the early detection of diseases in coffee crops using computer vision. Traditional crop monitoring is manual, costly, and prone to errors, resulting in significant productivity losses. The objective of this project was to develop and validate a SaaS platform that integrates a Deep Learning model for binary classification (healthy vs. unhealthy) of coffee leaves. The methodology involved the development of CNN models (highlighting the BinaryCNN_Deep architecture which achieved 90-95% accuracy), integrated into a web application with a map interface for field management. The results demonstrated that the solution is capable of identifying anomalies with high precision, offering producers an effective tool for rapid decision-making and waste reduction.

Keywords : Computer Vision; Coffee; Precision Agriculture; Deep Learning; Crop Monitoring.

List of Illustrations (optional item – NBR14724, item 4.2.1.9)

Figure 1 – [Figure 1 Title]..page]

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Figure 3 – [Figure 3 Title]..page]

Figure 4 – [Figure 4 Title]..page]

List of Tables (optional item – NBR14724, item 4.2.1.10)

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List of Abbreviations and Acronyms (optional item – NBR14724, item 4.2.1.11)

ACRONYM 1 Full name of acronym 1

ACRONYM 2 Full name of acronym 2

ACRONYM 3 Full name of acronym 3

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1 Introduction (For titles and numerical indicators, see NBR 14724, item 5.2.2; for progressive numbering, see NBR 6024)

The project presents CropTrack, an innovative solution for the agricultural sector, specifically focused on coffee farming. Brazil is the world's largest coffee producer, but the sector still faces significant challenges related to pest and disease management, which can cause losses of up to 30% in production if not managed correctly.

1.1 Context and Motivation:

The project operates at the intersection of Precision Agriculture (AgTech) and Computer Vision. It addresses the inefficiency of manual crop monitoring, where nutritional deficiencies and diseases often spread unnoticed until significant damage is done.

Large coffee producers manage vast areas where individual plant inspection is impossible. There is a market gap for automated, scalable solutions that can provide "eyes on the ground" without the labor cost of human patrols.

1.2 Problem Definition and Value Proposition:

Traditional monitoring relies on sampling or reactive measures (treating after damage is visible). This is slow, labor-intensive, and often inaccurate due to human error or fatigue.

CropTrack delivers a computer vision-based monitoring platform. By analyzing images of coffee leaves collected in the field, the system automatically classifies plant health (Healthy vs. Unhealthy). This allows for targeted intervention, reducing chemical usage and preventing crop loss.

1.3 Objectives of the Work:

Develop and validate a computational solution (SaaS platform) and a business plan for its market introduction.

Developed a Binary Classification Model (CNN) capable of distinguishing healthy coffee leaves from diseased ones with high accuracy (>90%).

Implement a Web Dashboard with map integration for field management and visual reporting.

Perform validation using real-world datasets and simulated field usage.

1.4 Justification and Contributions:

Coffee adds billions to Brazil's economy. Improving yield efficiency directly impacts national GDP.

The project demonstrates the practical application of Deep Learning (ResNet/EfficientNet architectures) in a resource-constrained environment (field usage), bridging the gap between academic research and commercial AgTech.

1.5 Work Structure:

The work is structured as follows: Chapter 2 details the Solution Development, including market analysis and technological implementation; Chapter 3 concludes the work.Section 2.4 details the technical architecture and AI models, while Section 2.5 specifically addresses the Business Plan.

2 [Solution Development]

2.1 [Definition of Market Assumptions and Hypotheses:]

[gotta fill this based on all the work that was done]

2.1.1 Problem Hypothesis

Large-scale coffee producers lose significant revenue due to the inefficiency of manual pest and disease monitoring. They face a lack of granular, real-time data to make localized treatment decisions.

2.1.2 Solution Hypothesis

A computer vision-based solution can accurately identify crop diseases from leaf images with sufficient speed and accuracy to replace or augment human inspection.

A map-based dashboard providing geolocation of diseased spots provides actionable intelligence that fits into the producer's workflow.

2.1.3 Value Hypothesis

Producers are willing to pay for a SaaS subscription that reduces input costs (pesticides) and increases yield, providing a clear Return on Investment (ROI).

2.2 [Market Sizing and Analysis:]

This subsection indicates the estimates of market size.

2.2.1 Market Size (TAM, SAM, SOM):

TAM (Total Addressable Market): The global precision farming market or the total value of Brazil's coffee production technology market.

Estimate: [INSERT TOTAL MARKET VALUE, e.g., $X Billion]

SAM (Serviceable Available Market): The segment of the Brazilian market focused on technological adoption in coffee (e.g., mechanized farms in Minas Gerais/São Paulo).

Estimate: [INSERT SAM VALUE, e.g., $Y Million]

SOM (Serviceable Obtainable Market): The realistic market share CropTrack can capture in the first 2-3 years, initially targeting early adopters in specific regions.

Estimate: [INSERT SOM VALUE, e.g., $Z Million]

2.2.2 Customer Segmentation and Profiling

"Roberto", 55 years old, owner of a medium-to-large coffee farm (>100 hectares). He values technology but requires solutions that are easy to integrate. He struggles with labor shortages for monitoring.

B2B, focusing on cooperatives and large independent producers in the Southeast region of Brazil.

2.3 [Competitive Analysis and Differentials:]

Analysis of the market and business environment.

Direct competition startups offering specific pest detection apps (e.g., Agrio, Plantix - though often B2C/generic).

Indirect competition Traditional drone monitoring services or satellite imagery providers.

Competitive Advantage:

Specific Focus: Optimized specifically for coffee pathologies common in Brazil.

Deep Learning Integration: Usage of state-of-the-art CNNs (BinaryCNN_Deep) with high validated accuracy (>90%).

Integration: Seamless connection between field data collection (app/upload) and visualization (dashboard).

2.4 [Technological Solution]

[This section should present the technological and computational solutions, which should include the following items:]

2.4.1 Requirements and Specifications:

Functional Requirements:

Allow users to draw field polygons on a map.

Upload images of crop leaves.

Classify images as "Healthy" or "Unhealthy" using proprietary models.

Display analysis results on the map with color-coded markers.

Non-Functional Requirements:

Accuracy of the model > 90%.

Response time < 5 seconds for inference.

Scalable backend architecture.

- User Specifications and Use Cases.

2.4.2 Architecture and Technology:

Client-Server architecture.

Frontend: React.js with Leaflet.js for map visualization.

Backend: Flask (Python) serving a REST API.

Database: SQLite (MVP) / PostgreSQL (Production).

AI Engine: PyTorch model (BinaryCNN) integrated into the backend.

2.4.3 Development and Implementation (MVP):

Agile development (Scrum) with 2-week sprints.

MVP Features: Field creation, Spot uploading, Binary Disease Classification, Heatmap visualization.

2.4.4 Testing and Technical Evaluation:

Testing Strategies: Unit testing for API endpoints, Manual testing for UI flows.

Results: The BinaryCNN_Deep model achieved 90-95% accuracy on the validation dataset (Sprint 2), proving robust enough for field usage.

2.5 [The Business Plan]

2.5.1 Market and Competitor Analysis:

Strengths: High accuracy specialized model, user-friendly interface.

Weaknesses: New brand in market, dependency on image quality.

Opportunities: Expansion to other crops (Soy, Corn), integration with drone imagery.

Threats: Climate unpredictability affecting client yield, entry of major agrochemical players into software.

- **Segmentation and Target Audience (Persona).**
- Analysis of Strengths, Weaknesses, Opportunities, and Threats **(SWOT Analysis)** .
- Analysis of **competitors** and product **differentials** .

2.5.2 Business Model (Business Model Canvas - BMC):

Value Proposition: Automated, precise health monitoring reducing operational costs.

Customer Segments: Medium-to-Large Coffee Producers, Cooperatives.

Revenue Streams: Monthly/Yearly Subscription (SaaS) per hectare.

Key Activities: Software Development, Model Retraining, Customer Success.

2.5.3 Marketing and Sales Strategy:

Partnerships with agricultural cooperatives and presence at AgTech events (e.g., Agrishow).

Continuous model updates and dedicated support.

2.5.4 Financial Projection and Feasibility:

Subscription tiers (Basic, Pro, Enterprise).

- Projected expenses, *break - even point* , and viability indicators (e.g. ROI).
- Definition of the initial **investment requirement** .

2.6 [Validation and Results]

[This section should be used to demonstrate the validation of the project in the market, which should not be merely a theoretical exercise.]

2.6.1 Validation Methodology:

- Description of the method used to test the **business hypothesis** and the **acceptance of the MVP** (e.g. , customer interviews, A/B testing, *landing page* with *lead generation*).

2.6.2 Market Validation Results:

- Presentation of collected data (Customer *feedback ,* Engagement metrics , Conversion rate, etc.).
- **Pivoting or Persisting:** Discussion about the changes (if any) made to the Business Model or Product based on real *feedback* .

2.6.3 Key Performance Indicators (KPIs):

- Presentation of key performance metrics (e.g. , *CAC* - Customer Acquisition Cost, *LTV* - Lifetime Value) . Value , *Churn Rate*).

2.6.4 Risks and Mitigation Plan:

- Identification of critical business risks (financial, technological, legal, competitive) and actions to mitigate them.

3 Conclusion

[In the conclusion, the author states whether or not the objectives set out in the project were achieved, presents future projections for the venture, and offers final considerations.]

References (mandatory item – NBR14724, item 4.2.3.1; NBR 6023)

(# References must comply with the rules of NBR6023 (Preparation of References;
Only the works consulted and cited in the text should be included in the list of references.
 # font size 12
 # prepared using single spacing;
 # aligned to the left margin of the text and without indentation;
 # separated from each other by a single blank line;
For online documents, the web address must be indicated, preceded by the expression "Available at:" and the access date preceded by the expression "Accessed on:";
The author should be indicated by their last name, in capital letters, followed by their first name and other last names, abbreviated or not; authors should be separated by semicolons.
When there are up to three authors, all must be listed;
When there are four or more authors, you can list all of them, or only the first one followed by the expression *et al* .

(# Examples:

Book:

[LAST NAME, First Name. **Book Title** : Subtitle. Edition. Publisher Location: Publisher, Publication Date.]

[PRESSMAN, RS; MAXIM, BR **Software Engineering** : A Practitioner's Approach. 9th ed. Porto Alegre: AMGH, 2021.]

[BASS, L.; CLEMENTS, P.; KAZMAN, R.; Software architecture in practice . 4th ed. Boston: Addison-Wesley, 2022.]

Article:

[LAST NAME, First Name. Article Title: Subtitle. **Journal Name** , Publication Location, Volume, Number, Pages, Publication Date. Available at: [electronic address]. Accessed on: [date of last access]]

[SANTOS, EH; BENITES, CS; STATERI, J. Study of artificial intelligence within the scope of PBL teaching. **Contemporary Journal** . [*S. l.*], v. 4, n. 9, p. 01-18, 2024. Available at:
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[VALENTE, JA; BITTENCOURT, II; SANTORO, FM; GARCIA, M.; ISOTANI, S.; GARCIA, A.; HABIMORAD, M. Project-Based Higher Education in Computing: Inteli on the path to innovation. **Brazilian Journal of Informatics in Education** , [*S. l.*], v. 33, p. 605–642, 2025. Available at:
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[VALENTE, JA *et al* . Project-Based Higher Education in Computing: Inteli on the path to innovation. **Brazilian Journal of Informatics in Education** , [*S. l.*], v. 33, p. 605–642, 2025. Available at:
https://journals-sol.sbc.org.br/index.php/rbie/article/view/4320. Accessed on: Oct. 27, 2025.]

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Appendices (optional item – NBR14724, item 4.2.3.3) (For titles without numerical indicators, see NBR 14724, item 5.2.3)

[Supporting document: text or document prepared by the author, used to support their argument and aid in understanding the main text without compromising the content presented in the body of the work. Ex : tables, reports, questionnaires, programming code, etc.]

Annexes (optional item – NBR14724, item 4.2.3.4)

[Supporting document: text or document not created by the author, but which serves as a basis, proof, or illustration of the content of the work. Ex : tables, reports, laws, manuals, etc.]